

Linear algebra – 2

Vector spaces, dimension and rank

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Theoretical minimum.

Definition. $\text{rank } A$ is the dimension of the column space; equivalently the largest size of a non-zero minor; equivalently the least r with $A = BC$, $B \in M_{n,r}$, $C \in M_{r,n}$; equivalently the least number of rank-one summands. It is unchanged by elementary row and column operations and by multiplication by an invertible matrix: $\text{rank}(AC) = \text{rank } A$ for invertible C .

Theorem 1 (the rank inequalities). For matrices of compatible sizes

$$\text{rank}(A + B) \leq \text{rank } A + \text{rank } B, \quad \text{rank}(AB) \leq \min(\text{rank } A, \text{rank } B),$$

$$\text{rank}(AB) \geq \text{rank } A + \text{rank } B - n \quad (\text{Sylvester}),$$

$$\text{rank}(AB) + \text{rank}(BC) \leq \text{rank } B + \text{rank}(ABC) \quad (\text{Frobenius}).$$

Equality in Sylvester's inequality holds exactly when $\ker A \subseteq \text{im } B$, and $\text{rank}(A | B) \leq \text{rank } A + \text{rank } B$ for the block row.

Theorem 2 (dimension counting). $\dim \ker A + \text{rank } A = n$; for subspaces $\dim(U + W) + \dim(U \cap W) = \dim U + \dim W$. A functional vanishing on $\ker f_1 \cap \dots \cap \ker f_n$ is a linear combination of $f_1, \dots, f_n$ — the dual form of the same count, and the reason why “a linear condition follows from linear conditions” always means “is their combination”.

Fact 1 (transposes and real matrices). $\text{rank } A = \text{rank } A^T = \text{rank}(A^T A)$ over $\mathbb{R}$, but not over $\mathbb{C}$ or over $\mathbb{F}_2$; the Moore–Penrose pseudoinverse satisfies $AA^+A = A$ and $\text{rank } A^+ = \text{rank } A$.

Fact 2 (dimension as a counting tool). To bound the number of objects, place them in a vector space and prove independence; to prove that objects

exist, count dimensions and use that a subspace of dimension > 0 contains a non-zero vector. A rank-one decomposition $A = \sum_{i=1}^r u_i v_i^\top$ turns a statement about a matrix into a statement about r vectors, which is why the rank is the right measure of complexity.

1. Prove the inequalities of Theorem 1, show that Sylvester's and Frobenius' bounds are sharp for every admissible triple of ranks, and prove that equality in Sylvester's inequality holds exactly when $\ker A \subseteq \operatorname{im} B$.
2. (IMC 1998) Let V be a vector space and let $f, f_1, f_2, \dots, f_n$ be linear maps from V to $\mathbb{R}$. Suppose that $f(x) = 0$ whenever $f_1(x) = f_2(x) = \dots = f_n(x) = 0$. Prove that f is a linear combination of $f_1, f_2, \dots, f_n$.
3. (Putnam 2009) Say that a polynomial with real coefficients in two variables x, y is balanced if the average value of the polynomial on each circle centred at the origin is 0. The balanced polynomials of degree at most 2009 form a vector space V over $\mathbb{R}$. Find the dimension of V .
4. (IMC 2005) Let A be the $n \times n$ matrix whose i, j entry is $i + j$ for all $i, j = 1, 2, \dots, n$. What is the rank of A ?
5. (Romania national olympiad 2001) Let $A \in M_n(\mathbb{C})$ have rank r , where $n \geq 2$ and $1 \leq r \leq n - 1$.
 - a) Show that there are $B \in M_{n,r}(\mathbb{C})$ and $C \in M_{r,n}(\mathbb{C})$ of rank r with $A = BC$.
 - b) Show that A satisfies a polynomial equation of degree $r + 1$ with complex coefficients.
6. (ФКН, осень 2022) Let $A, B \in M_n(\mathbb{R})$, $n \geq 2$, with $B^2 = B$. Prove that $\operatorname{rank}(AB - BA) \leq \operatorname{rank}(AB + BA)$.
7. (SEEMOUS 2008) Find all surjective functions $f: M_n(\mathbb{R}) \rightarrow \{0, 1, \dots, n\}$ such that $f(XY) \leq \min\{f(X), f(Y)\}$ for all $X, Y \in M_n(\mathbb{R})$.

8. (IMC 2022) Let $A_1, A_2, \dots, A_k$ be $n \times n$ idempotent complex matrices such that $A_i A_j = -A_j A_i$ for all $i \neq j$. Prove that at least one of them has rank at most $\frac{n}{k}$.